Leo Shaw

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Education

Carnegie Mellon University *Pittsburgh, PA*
Master of Science in Artificial Intelligence Engineering - Mechanical Engineering *Dec 2026*
Rutgers University *New Brunswick, NJ*
Bachelor of Science in Mechanical Engineering; Bachelor of Arts in Computer Science *Dec 2024*

Skills

Programming/AI: Python, C++, Java, PyTorch, TensorFlow, OpenCV, YOLO, Stable-Baselines3
Design/Analysis: SOLIDWORKS, ANSYS, MATLAB
Hardware: PLC/HMI integration, circuitry, soldering, 3D printing, mills, lathes

Experience

Field Service Engineer I *Santa Clara, CA / Ulvac Technologies, Oct 2024 - Aug 2025*
- Rebuilt and installed an upgraded 8-inch etch chuck assembly, integrated a new VAT system and circuit breaker, and modified valve-control software and wiring, improving tool flexibility, reducing unplanned downtime by 15%, and increasing wafer temperature uniformity.
- Maintained semiconductor tools across 6 high-volume fabs through preventive maintenance, calibration, and data-driven repairs, increasing uptime by 18%.
- Executed 30+ hardware and software retrofit projects across vacuum, RF, motion-handling, PLC, and HMI subsystems, improving process stability by 22% and extending tool life.
- Led root-cause analysis with engineering and manufacturing teams, implemented corrective actions, and standardized troubleshooting procedures, reducing average fault diagnosis time by 35%.

Mechanical Engineering Intern *Milpitas, CA / Paltorc, May 2022 - Aug 2022*
- Integrated the E-Bike control module with a mobile app to stream heart rate, temperature, weather, and incline data in real time.
- Diagnosed and resolved mechanical and software issues to improve system reliability.

Research & Projects

Graduate Research Assistant *Pittsburgh, PA / Carnegie Mellon University, May 2026 - Present*
- Developing small language model workflows for manufacturing process design, focusing on reliable expert-agent pipelines, retrieval, and deployment tradeoffs.
- Curating and preprocessing manufacturing data for thin-film deposition, physical vapor deposition, and sputtering workflows to support downstream model training and analysis.

Autonomous Driving Local Planner with Reinforcement Learning *Jan 2026 - May 2026*
- Built a closed-loop CARLA planning stack combining A*-based global routing, PPO local trajectory generation, and fixed PID control for highway merge and unprotected left-turn scenarios.
- Developed the training and evaluation pipeline in Python using Stable-Baselines3, PyTorch, and CARLA, including scenario generation, reward shaping, route-relative observations, and rollout analysis.
- Improved highway-merge PPO success from 76% to 83% while reducing 90th-percentile completion time from 24.5 s to 15.25 s.
- Improved left-turn success from 44% to 80% in closed-loop evaluation, reaching performance comparable to the rule-based baseline.

Real-Time Conveyor-Belt Object Detection and Robotic Sorting *Jan 2026 - May 2026*
- Developed a computer vision and robotic sorting pipeline for bottles, cans, and optional six-pack bundles using Python, YOLO/OpenCV, homography-based coordinate mapping, inverse kinematics, and delta-arm motion control.
- Improved localization by refining centroids within detections, projecting image coordinates into robot-plane coordinates, and compensating for conveyor motion and inference latency during pick timing.
- Integrated distributed perception and actuation across a Raspberry Pi, cloud vision pipeline, servo-driven delta arm, and pneumatic vacuum end effector to support closed-loop pick-and-place operation.
- Achieved up to 99% classification accuracy and about 50% physical sorting accuracy in the integrated system; key limitations were vision latency, conveyor speed variation, and structural compliance.

Solar-Powered Terrain Walker *Aug 2023 - May 2024*
- Designed and built an autonomous solar-powered quadruped using Chebyshev 4-bar linkages and pantographs, reducing cost by 34% versus comparable 6-legged designs.
- Modeled pivot mechanisms in SOLIDWORKS, validated structural integrity in ANSYS, and implemented motion and sensing subsystems in C on Arduino.